

Identification and Control of a Propeller-Actuated Damped Pendulum (PADP)

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Abstract—This paper presents the experimental identification and closed-loop control of a propeller-actuated damped pendulum (PADP). Three step-response datasets were collected and processed using both manual parameter approximation and MATLAB’s System Identification toolbox, yielding six candidate transfer functions. Statistical analysis identified one experiment as disturbance-affected and excluded it; the final plant model was obtained by averaging the four remaining models. Three continuous-time controllers were designed — a PID tuned via IMC in SISOTool, an integral state feedback regulator via LQR (ISF-LQR), and an I-PD by pole placement — and physically validated against a settling time of 5 s and maximum overshoot of 3%. The ISF-LQR and I-PD controllers satisfied both specifications, while the PID exhibited the poorest transient response among the three.

I. Introduction

The propeller-actuated damped pendulum (PADP) is an underactuated mechanical system whose angular position is governed by the aerodynamic torque of a brushless motor and propeller assembly, opposing gravity and bearing damping. Around a fixed operating point the nonlinear dynamics admit a linear second-order underdamped approximation [1], making it a compact yet representative platform for comparing identification and control strategies.

Accurate plant identification is the prerequisite for any model-based controller. Parametric methods extract the natural frequency ω_n and damping ratio ζ directly from measured transient features such as peak overshoot and oscillation period [2]. Data-driven tools, such as MATLAB’s ident toolbox, complement this by fitting transfer function prototypes to minimize a prediction-error criterion. Because experimental data are always subject to noise and occasional disturbances, statistical screening across repeated trials is essential before any model averaging is performed.

Given a validated plant model, different control architectures offer distinct trade-offs. The PID with Internal Model Control (IMC) tuning [3] is widely used for its simplicity, but its derivative action on the error signal can amplify noise and produce overshoot in oscillatory plants. The Linear Quadratic Regulator with an augmented integral state (ISF-LQR) provides a systematic, optimization-based synthesis that simultaneously penalizes control effort and tracking error, and guarantees zero steady-state

offset [4]. The I-PD structure — where only the integrator acts on the error while the proportional and derivative terms act exclusively on the plant output — eliminates derivative kick and typically reduces overshoot compared to the parallel PID [5]; pole placement then allows the designer to prescribe closed-loop dynamics directly from the time-domain specifications.

This paper reports the complete identification-to-validation cycle for the PADP. Section II covers data acquisition, parameter estimation, statistical model selection, and the derivation of the final plant. Section III details the three controller designs. Sections IV and V present and interpret the experimental results against the design requirements of settling time $t_s \leq 5$ s and maximum overshoot $M_p \leq 3\%$.

II. System Identification

A. Experimental Setup and Data Acquisition

The PADP platform consists of an extruded-aluminium support rigidly fixed to a wooden base. A hollow aluminium arm pivots on bearings at the top of the support and carries a Hobbymate 2204 brushless motor governed by a SimonK 20 A ESC fitted with a GEMFAN 5045BNR three-blade propeller. Angular position is measured by a USDigital E5 incremental encoder (3600 CPR) coupled to the pivot shaft and sampled every 20 ms with a gain $k = 2\pi/3600$ rad/count. PWM actuation uses OneShot125 mode at a 500 μ s period. Three independent experiments were conducted by applying a step input of 1.7 units from rest, establishing a steady-state operating point near $\theta_{eq} \approx 0.6$ rad ($\approx 34.4^\circ$), and recording the angular response for 40 s.

B. Parameter Approximation

The logarithmic decrement method was applied to each trial to extract ω_n , ζ , and the static gain K from the measured transient features (successive peak amplitudes and oscillation period):

$$\delta = \frac{\ln(y_0/y_i)}{i}, \quad \zeta = \frac{\delta}{\sqrt{(2\pi)^2 + \delta^2}}, \quad \omega_n = \frac{2\pi/T}{\sqrt{1 - \zeta^2}} \quad (1)$$

yielding three second-order transfer functions of the prototype:

$$G_i(s) = K_i \frac{\omega_{n,i}^2}{s^2 + 2\zeta_i \omega_{n,i} s + \omega_{n,i}^2}, \quad i = 1, 2, 3 \quad (2)$$

TABLE I: Parameters estimated by the logarithmic decrement method

Trial	K	ζ	ω_n (rad/s)
1	0.338	0.032	3.9292
2	0.340	0.035	3.9294
3	0.423	0.033	3.9291

C. MATLAB System Identification

MATLAB's ident toolbox was used in parallel to fit the same second-order prototype to each dataset via prediction-error minimization, producing three additional transfer functions with their respective best-fit percentages.

TABLE II: Parameters estimated by system identification

Trial	K	ζ	ω_n (rad/s)
1	0.346	0.0292	3.9225
2	0.346	0.0295	3.9301
3	0.426	0.0296	3.9091

D. Statistical Analysis and Model Selection

The six identified models were pooled and their parameters compared statistically. The mean $\bar{x}$ and sample standard deviation s were computed for K , ζ , and ω_n . The two models derived from trial 3 showed parameter deviations exceeding $2s$ from the group mean, consistent with an external disturbance during that recording session. Those two models were therefore excluded, and the final plant was obtained by averaging the parameters of the four remaining transfer functions:

$$G(s) = \frac{K_{\text{avg}} \omega_{n,\text{avg}}^2}{s^2 + 2\zeta_{\text{avg}} \omega_{n,\text{avg}} s + \omega_{n,\text{avg}}^2} \quad (3)$$

$$G(s) = \frac{5.287}{s^2 + 0.2481s + 15.43} \quad (4)$$

III. Controller Design

Design specifications for all regulators: settling time $t_s \leq 5$ s (2% criterion) and maximum overshoot $M_p \leq 3\%$ to a 0.6 rad step reference, with zero steady-state error and disturbance rejection.

A. Continuous PID Controller (SISOtool)

A continuous-time PID controller was designed for the identified second-order plant model using the Internal Model Control (IMC) tuning methodology. Although the imposed specification allows up to 3% overshoot, the IMC tuning parameter was deliberately tightened until the simulated closed-loop response exhibited $\%OS \approx 0\%$.

This conservative choice was motivated by the physical behaviour of the plant: the PAMH already introduces an inherent overshoot at the process level, so additional overshoot contributed by the feedback controller would accumulate, potentially violating the original specification in real operating conditions. By driving the controller-induced overshoot to zero, the closed-loop response remains within the 3% bound even when the plant's natural overshoot is taken into account.

The implemented controller incorporates a filtered derivative action to attenuate high-frequency noise amplification. Its transfer function is given by ??

$$PID(s) = K_p + \frac{K_i}{s} + K_d s \cdot \frac{N}{s + N} \quad (5)$$

where K_p is the proportional gain, K_i the integral gain, K_d the derivative gain and N the derivative filter coefficient.

Once the IMC-equivalent open-loop transfer function $C(s)G(s)$ was obtained, its numerator and denominator polynomials were extracted in MATLAB. Identifying the coefficients of (5) with those polynomials, and setting N equal to the second denominator coefficient. The tuning procedure yielded the following parameter values:

TABLE III: IMC-tuned PID controller parameters.

K_p	K_i	K_d	N
-0.634	2.000	0.343	3

The negative proportional gain is consistent with the sign convention of the identified plant, which exhibits an inherently inverting static gain.

The PID controller with their values can be written as:

$$C(s) = -0.634 + \frac{2}{s} + 0.343 \frac{3s}{s + 3} \quad (6)$$

B. Integral State Feedback Controller (ISF) via LQR

An Integral State Feedback (ISF) controller was designed using the Linear Quadratic Regulator (LQR) criterion. The method augments the plant with an integral state to ensure zero steady-state error and improve disturbance rejection.

The identified plant in controllable canonical form is:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} -0.2481 & -15.4300 \\ 1 & 0 \end{bmatrix}, & \mathbf{B} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ \mathbf{C} &= [0 \quad 5.2870], & \mathbf{D} &= 0 \end{aligned} \quad (7)$$

The system is fully controllable (rank = 2). To include integral action, the augmented model is defined as:

$$\dot{x}_I = r - \mathbf{C}\mathbf{x} \quad (8)$$

$$\mathbf{A}_a = \begin{bmatrix} -0.2481 & -15.4300 & 0 \\ 1 & 0 & 0 \\ 0 & -5.2870 & 0 \end{bmatrix}, \quad \mathbf{B}_a = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (9)$$

The augmented system is also controllable (rank = 3). Based on the specifications ($t_s \leq 5$ s, $M_p \leq 3\%$), the desired poles were selected as:

$$p_{1,2} = -2.8 \pm 3.1266j, \quad p_3 = -14.0 \quad (10)$$

The LQR weighting matrices were chosen using Bryson's rule:

$$\mathbf{Q} = \text{diag}\left(\frac{1}{9^2}, \frac{1}{7^2}, \frac{1}{0.45^2}\right), \quad R = \frac{1}{0.6^2} \quad (11)$$

The resulting gain vector $\hat{\mathbf{K}} = [k_1 \ k_2 \ K_i]$ is:

$$\hat{\mathbf{K}} = [0.7214 \ 0.4370 \ -1.3333] \quad (12)$$

which yields the closed-loop poles:

$$p_{1,2} = -0.2593 \pm 3.9454j, \quad p_3 = -0.4509 \quad (13)$$

C. I-PD Controller via Pole Placement

The I-PD structure places the integrator on the error signal while the proportional and derivative terms act on the plant output, preventing derivative kick on step changes of the reference:

$$C_{\text{I-PD}}(s) = \frac{K_i}{s} - (K_p + K_d s) \quad (14)$$

The target specifications ($t_s \leq 5$ s, $M_p \leq 3\%$) were mapped to the damping ratio ζ_d and natural frequency $\omega_{n,d}$ of the desired dominant poles. A third real pole was placed sufficiently far to the left to remain non-dominant. Coefficient matching of the closed-loop characteristic polynomial yielded K_p , K_d , and K_i , subsequently decomposed into P, I, D, N parameters for the PASCAL PID block. The controller was designed to specifications more strict than the nominal requirements ($t_s \leq 2.9$ s, $M_p \leq 1\%$) to provide a performance margin that compensates for the degradation expected during physical implementation.

$$p_{1,2} = -1.3793 \pm j0.4705 \quad p_3 = -6.8966$$

$$K_p = 1.0817, \quad K_i = 2.7704, \quad K_d = 1.7793, \quad N = 20$$

The third real pole p_3 was placed at five times the magnitude of the real part of the dominant poles, following the standard non-dominance criterion [1]:

$$|p_3| = 5 |\text{Re}(p_{1,2})| = 5 \times 1.3793 = 6.897$$

This placement ensures that the transient contribution of p_3 decays approximately five times faster than the dominant pair

D. Discrete PID Controller (SISOtool)

A discrete-time PID controller was designed through manual pole-zero placement in the z -domain, without resorting to an automated tuning method such as IMC. The compensator structure was chosen to consist of two real poles and two complex conjugate zeros, which corresponds to the discrete equivalent of a PID action with filtered derivative. The design procedure consisted of iteratively adjusting the controller gain, the natural frequency associated with the complex zero pair, and the pole locations until the closed-loop step response satisfied the specifications $T_s < 5$ s and $\%OS < 3\%$.

The two poles were placed at $z = 1$, which introduces pure integral action and ensures zero steady-state error to a step reference, and at $z = 0.09$, which provides the derivative filtering role analogous to the N parameter in the continuous design. The two complex zeros were positioned by tuning their associated natural frequency and damping ratio to shape the transient response, adding phase lead in the frequency band relevant to the settling requirement. The overall controller gain was then adjusted to achieve the desired closed-loop speed of response while keeping the overshoot within the imposed bound.

The resulting discrete controller transfer function is:

$$C(z) = \frac{3.0189(z^2 - 1.95z + 0.9881)}{(z - 0.09)(z - 1)}, \quad (15)$$

IV. Results and Analysis

This section presents the results obtained in both simulation and physical implementation for each controller, evaluating compliance with the performance specifications ($t_s \leq 5$ s, $M_p \leq 3\%$).

A. Continuous PID Controller

The figure 1 shows the simulated closed-loop step response obtained by connecting the designed controller around the identified plant model. The output settles within the $\pm 2\%$ band with the response oscillating persistently around the reference once inside it. The sustained oscillation indicates that the simulation is operating at the boundary of the stability margin introduced by the IMC tuning.

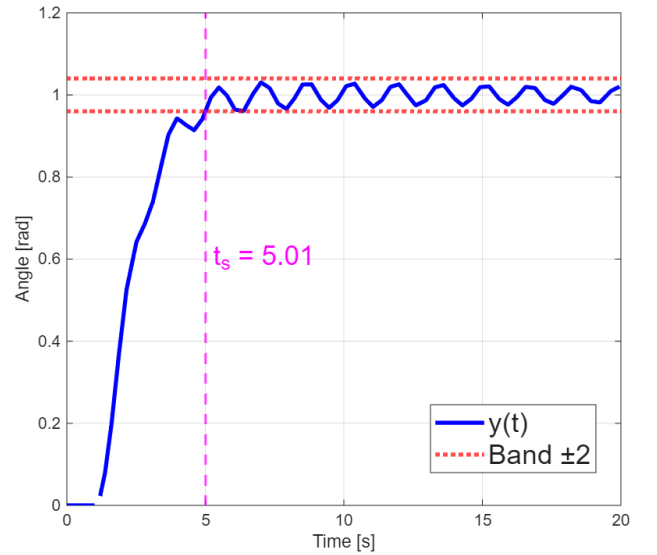


Fig. 1: Simulated step response of the PID controller.

The experimental step response recorded on the physical platform. The output enters and remains inside the $\pm 2\%$ band with a small peak overshoot before settling cleanly, as summarised in Table IV. A disturbance injected at

$t = 20$ s causes a transient excursion, after which the controller successfully rejects it and returns the output to the reference, demonstrating integral action and robustness to step disturbances.

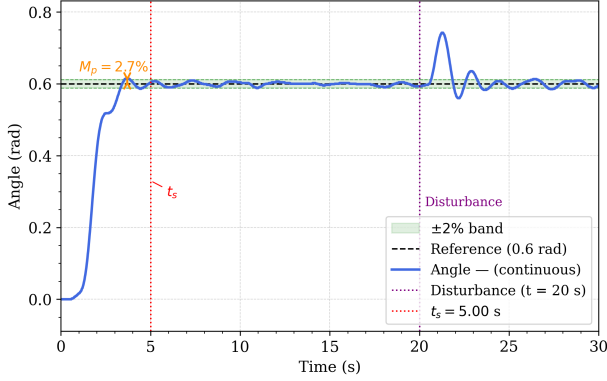


Fig. 2: Experimental step response of the PID controller.

The primary difference between the two responses lies in the steady-state behaviour after settling. In simulation, the output exhibits a sustained low-amplitude oscillation that keeps it cycling inside the $\pm 2\%$ band, whereas experimentally the response damps out and converges to the reference without visible oscillation. Furthermore, the experimental response presents a visible peak overshoot before entering the band, a feature absent in the simulated response.

To improve the experimental behaviour, the derivative filter coefficient was adjusted from $N = 3$ to $N = 5$ while keeping K_p , K_i , and K_d unchanged. Increasing N raises the filter cutoff frequency, allowing the derivative action to respond over a wider frequency range. This produces a stronger damping effect on the real platform, which explains both the suppression of the post-settling oscillation and the cleaner convergence observed experimentally. In simulation, the linear model does not capture the higher-frequency dynamics and sensor noise present on the physical system, so the lower value of N was sufficient to meet the band criterion, albeit with residual oscillation. On the real platform, the increased derivative activity provided by $N = 5$ attenuates those higher-frequency components, resulting in a smoother and more stable steady-state response.

TABLE IV: PID performance metrics: simulation vs. experiment

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	5.01	3.4	0	No
Experimental	5	2.7	0	Yes

B. Integral State Feedback Controller via LQR

The closed-loop step responses of the ISF-LQR controller for $r = 0.6$ rad are shown in Fig. 1 (experiment)

and Fig. 2 (simulation). The corresponding performance metrics are summarized in Table V.

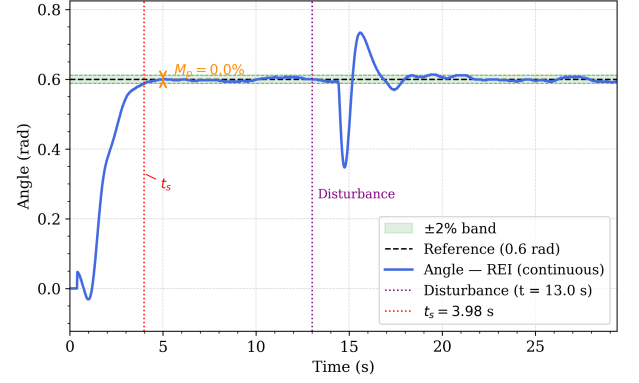


Fig. 3: Experimental step response of the ISF-LQR controller.

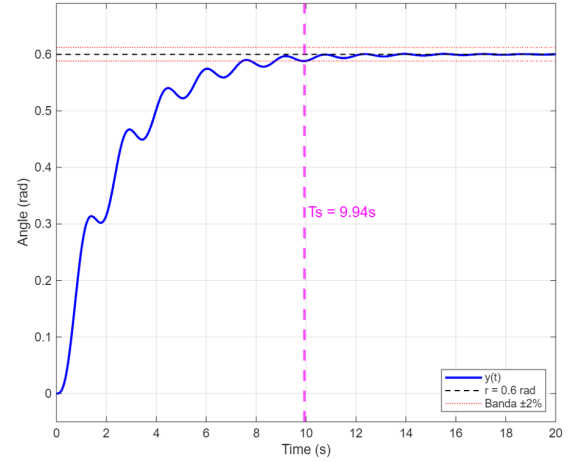


Fig. 4: Simulated step response of the ISF-LQR controller.

TABLE V: ISF-LQR performance metrics: simulation vs. experiment

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	9.94	0.12	0	No
Experimental	3.98	0	0	Yes

As observed, the simulated response exhibits a settling time of 9.94 s, exceeding the design requirement of 5 s, although the overshoot remains negligible at 0.12%. In contrast, the experimental implementation achieves a significantly faster settling time of 3.98 s with no observable overshoot, fully complying with the specified performance criteria. Both cases satisfy the steady-state error requirement ($e_{ss} = 0$) due to the inclusion of integral action. Additionally, the overshoot constraint ($M_p \leq 3\%$) is met in both simulation and experimental results.

C. I-PD Controller via Pole Placement

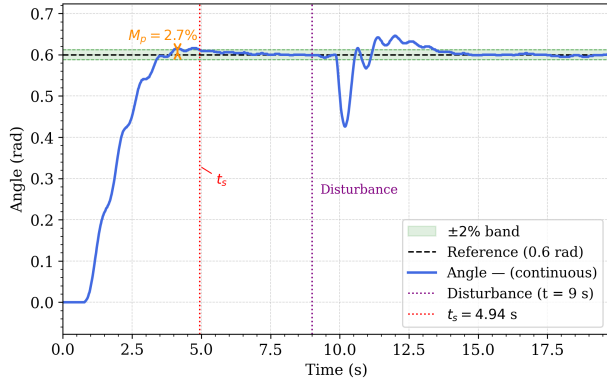


Fig. 5: Experimental step response of the ISF-LQR controller.

D. Discrete PID Controller

Although the discrete PID controller (15) was designed through careful manual pole-zero placement, its experimental implementation on the physical plant did not yield a satisfactory closed-loop response. The controller failed to meet the imposed specifications under real operating conditions, exhibiting behaviour that deviated significantly from the expected performance. This outcome is attributed to the sensitivity of the manual tuning approach to unmodelled dynamics present in the physical system, which are not reflected in the linear model used during the design stage.

In light of these results, neither the simulation nor the experimental step response plots are included for this controller, as the experimental failure renders a comparative analysis inconclusive. The discrete PID design is documented here for completeness, to illustrate the limitations of manual frequency-domain tuning when unmodelled dynamics play a significant role in the closed-loop behaviour.

E. Comparative Summary

V. Discussion

This section interprets the results presented above, analyzing the causes of the observed differences between theoretical models and the physical behavior of the system, as well as the advantages and disadvantages of each implemented control strategy.

A. ISF-LQR: Simulation–Experiment Discrepancy

Figs. 4 and 5 show a clear contrast between simulated and experimental responses. The simulation exhibits a slowly damped transient with $t_s = 9.94$ s, consistent with the real pole $p_3 = -0.4509$ near the imaginary axis, which dominates the settling behavior despite the faster complex poles. In contrast, the experimental response converges monotonically to the 0.6 rad reference in $t_s = 3.98$ s with zero overshoot, satisfying both specifications.

A small negative deflection at $t \approx 1$ s in the experimental response (about -0.07 rad) is attributed to actuator dead-zone and propeller inertia, which delay the onset of effective control action. This effect is not captured by the linear model.

The faster experimental convergence is consistent with additional nonlinear damping (e.g., friction and aerodynamic drag), which increases the effective damping and suppresses oscillations, leading to a shorter settling time than predicted.

Finally, the experimental response shows effective disturbance rejection; a perturbation at $t \approx 13$ s produces a deviation of approximately 0.25 rad, after which the output returns to the reference, remaining within the margin of less than 2% steady-state error. This confirms the role of comprehensive action in rejecting disturbances despite the model’s inaccuracies.

VI. Conclusions

The ISF-LQR controller achieved zero steady-state error and effective disturbance rejection. While the simulation resulted in $t_s = 9.94$ s and $M_p = 0.12\%$, failing the settling time requirement, the experimental response reached $t_s = 3.98$ s with $M_p = 0\%$, satisfying all specifications.

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